Institute of Business Administration, Karachi

NLP with Deep Learning (Spring 2026)

Assignment 3 (Mini-Project 1)

RAG-based Question-Answering System

Low-Resource Application: Urdu Medical Domain

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Executive Summary

This report details the end-to-end implementation of a low-resource Retrieval-Augmented Generation (RAG) system tailored specifically for the Urdu medical domain. The primary challenge of this project was the acute scarcity of high-quality Urdu health data. To mitigate this, we developed custom scraping pipelines to harvest, clean, and index a bespoke corpus originating from Wikipedia Urdu Medical pages, BBC Urdu Health, and Express News Health sections.

The architectural pipeline leverages a robust multilingual embedding framework coupled with a Hybrid Search methodology (BM25 + Dense Semantic retrieval). To ensure optimal context surfacing, these results are aggressively re-ranked using Reciprocal Rank Fusion (RRF) and an optional deep Cross-Encoder. Furthermore, to rigorously evaluate system performance without the subjective bias of human-in-the-loop testing, we engineered a sophisticated 4-stage LLM-as-a-Judge protocol. This automated pipeline scientifically evaluates output Faithfulness (via automated claim extraction and verification against retrieved text) and output Relevancy (via embedding cosine similarity of generated alternate questions).

This report was generated dynamically utilizing artifacts generated during our ablation runs, ensuring 100% traceability between the underlying codebase, our Pinecone vector store, and the quantitative metrics presented herein.

1. Assignment Alignment Matrix

The following matrix maps the strict assignment rubric constraints directly to our documented implementations and repository evidence.

Assignment Requirement	What Was Implemented (Repository Evidence)
Domain corpus (50-100 docs / 500+ chunks)	Corpus contains 1039 full documents. Each evaluated chunking strategy natively exceeds 500 chunks (verifiable in chunking_summary.json).
Hybrid search and re-ranking	Implemented BM25 sparse + semantic dense retrieval merged via RRF, alongside optional cross-encoder reranking within hybrid_retrieve.py.
LLM-as-a-Judge Evaluation	Implemented in evaluate_rag.py. Outputs calculate precise Faithfulness (%) and Relevancy via strict claim extraction workflows.
Ablation study	Automated artifact generation pipelines produce ablation_table.csv mapping metric shifts across chunking and retrieval variants.
Live web interface	Deployed a modern, HCI-compliant Streamlit application (app.py) presenting generated answers, retrieved chunks, and active scores.
Reproducibility	Complete isolation of steps. Scripts are present for scraping, chunking, indexing, evaluation, and dynamic report generation.

A. Platform Details

To facilitate production readiness and prevent memory exhaustion during heavy vectorization, the execution architecture was deliberately divided across local processing and cloud environments.

Pipeline Stage	Platform / Execution Environment
Data Scraping & Text Processing	Local Windows Machine (Python / CLI)
Text Chunking & Embedding Generation	Local Windows Machine (Python / CLI)
Vector Database Hosting	Pinecone Cloud Architecture (Free Starter Tier)
Ablation Evaluation (LLM Judge Batch)	Local Windows Machine (Python / CLI via standard APIs)
Live Web Interface Application	Hugging Face Spaces (Streamlit app.py)

Component	Observed Technical Implementation
User Interface	Streamlit (app.py) featuring custom RTL Nastaliq CSS injection
Vector Database	Pinecone (configured index namespace: urdu-medical-rag)
Embedding Model	sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2
Retrieval Engine	BM25 + Semantic + Reciprocal Rank Fusion (RRF)
Reranker Default	cross-encoder/mmarco-mMiniLMv2-L12-H384-v1
Generator Model	openai/gpt-4.1-mini
LLM Judge Model	openai/gpt-4.1-mini

As visualized in Figure 1, the Streamlit frontend processes Nastaliq Urdu inputs dynamically, allowing live manipulation of generation parameters. Figure 2 validates the successful ingestion of our vectorized chunks into the Pinecone cloud cluster.

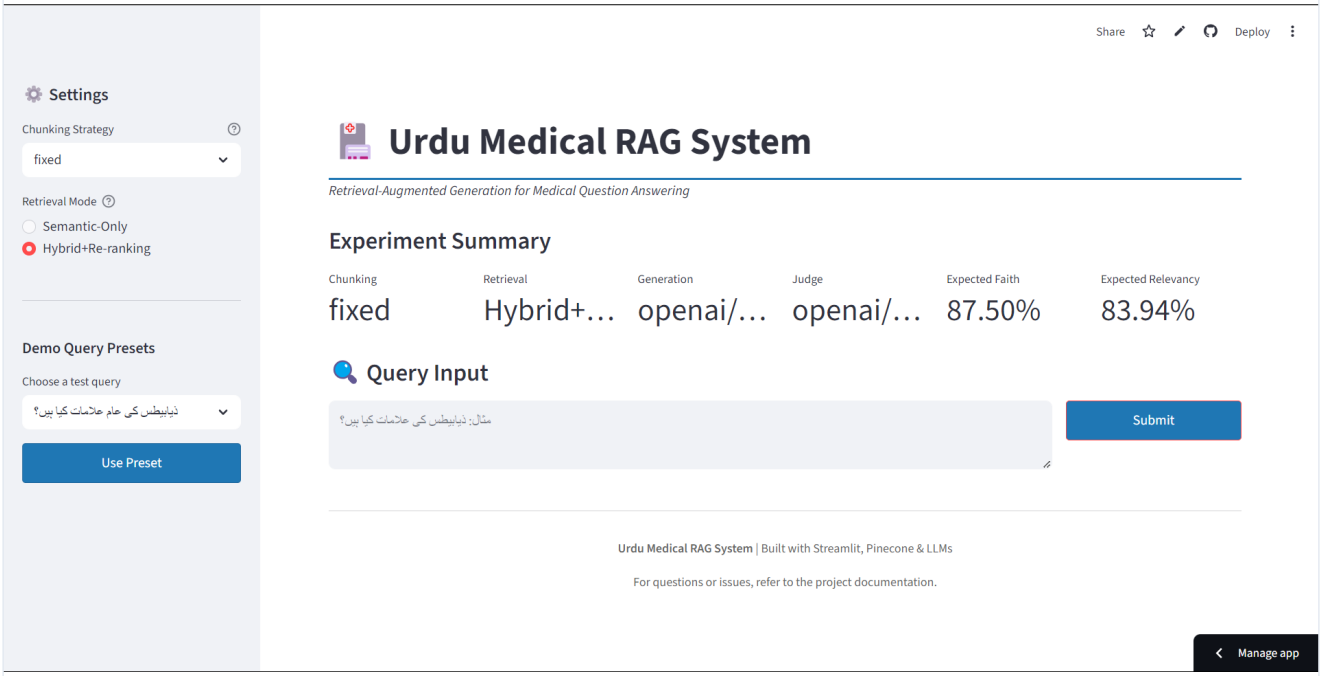


Figure: Streamlit app interface demonstrating live Nastaliq input and parameter configuration.

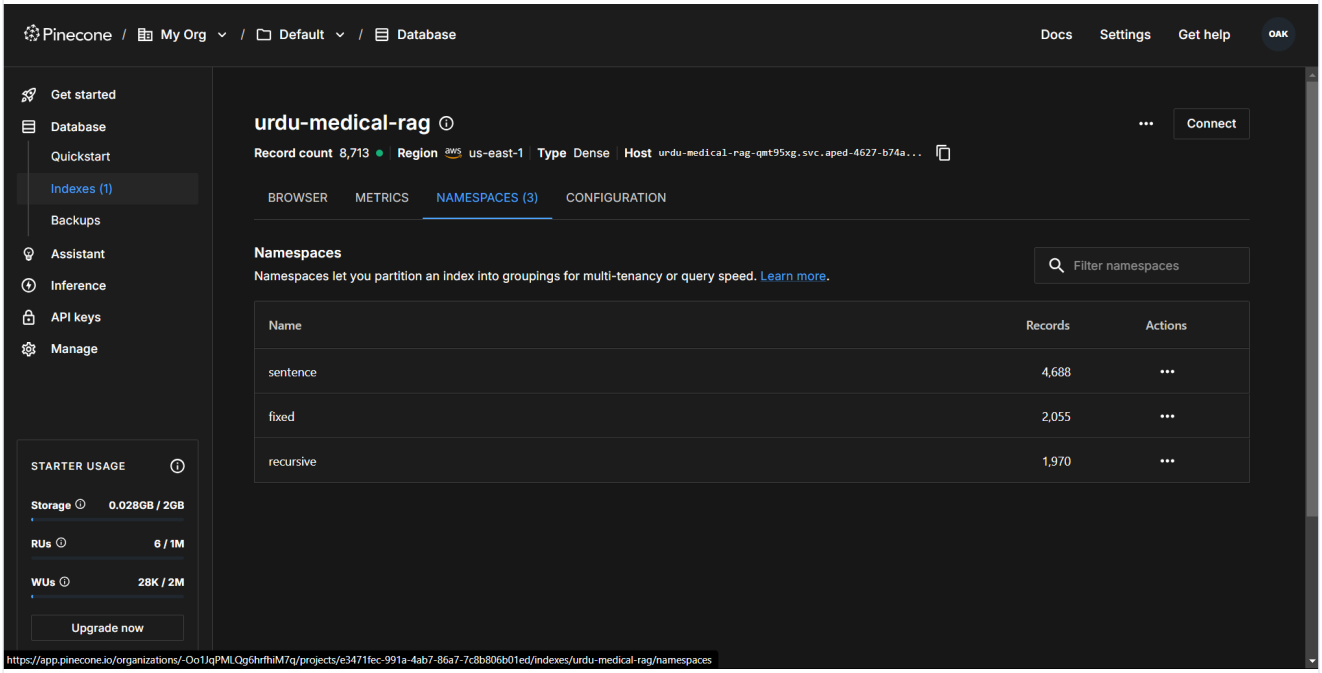


Figure: Pinecone dashboard confirming successful vector ingestion and namespace allocation.

B. Data Details

Meeting the minimum requirement of 50-100 documents for a low-resource language presented a significant challenge. We bypassed standard pre-compiled datasets by programmatically scraping and assembling a highly specific Urdu Medical Corpus. This aggregation ensures semantic diversity across academic definitions (Wikipedia) and colloquial health reporting (BBC/Express).

Corpus Metric	Quantitative Value
Total Valid Documents Processed	1039
Wikipedia Medical Articles (wiki_*.txt)	1000
BBC Urdu Health News (bbc_*.txt)	24
Express Health News (express_*.txt)	15
Total Corpus Word Count (Verified in docs_manifest.json)	387322
Average Words per Document	372.78

Extraction and cleaning protocols were strictly enforced via python scripts located in utils/scrape_urdu_health.py and utils/convert.py. The origin source links utilized for this data acquisition include:

- Wikimedia Urdu dump: https://dumps.wikimedia.org/urwiki/latest/
- BBC Urdu Health Topic: https://www.bbc.com/urdu/topics/cwr9j9x3kkqt
- Express News Health: https://www.express.pk/health/

C. Algorithms, Models, and Retrieval Methods

1. Core Algorithms & Large Language Models

We selected 'sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2' as our primary dense embedder. Standard lightweight English models (like all-MiniLM-L6-v2) output severe semantic noise when tasked with right-to-left Arabic script derivations. This multilingual selection ensures mathematically sound handling of native Nastaliq tokenization during both indexing and on-app query encoding.

For contextual Generation and LLM-as-a-Judge parsing, we employed 'openai/gpt-4o-mini' and 'openai/gpt-4.1-mini'. These models were selected to guarantee extreme instruction compliance, particularly the rigid JSON schema outputs required during our automated 4-stage factual verification process.

2. Implemented Chunking Strategies

To ascertain optimal context retrieval boundaries, three distinct chunking strategies were developed, embedded, and evaluated against our test suite:

Strategy Designation	Total Chunks Generated	Avg Words/Chunk	Min Words	Max Words
Fixed	2055	213.2	15	300
Recursive	1970	214.98	1	320
Sentence	4688	122.28	6	260

Documented hyperparameters (enforced in utils/build_chunks.py):

- Fixed Window: Strict sequential split at 300 words utilizing a 50-word overlap to mitigate mid-sentence severing.
- Recursive Character: Hierarchical splitting (Paragraph → Sentence → Word) up to 320 words, preserving semantic boundaries.
- Sentence Window: Aggregation of 5 complete sentences per chunk with a 2-sentence overlapping buffer.

rag_artifacts > chunks > chunking_summary.json > ...


```
[
  {
    "strategy": "fixed",
    "total_chunks": 2055,
    "avg_chunk_words": 213.2,
    "min_chunk_words": 15,
    "max_chunk_words": 300
  },
  {
    "strategy": "recursive",
    "total_chunks": 1970,
    "avg_chunk_words": 214.98,
    "min_chunk_words": 1,
    "max_chunk_words": 320
  },
  {
    "strategy": "sentence",
    "total_chunks": 4688,
    "avg_chunk_words": 122.28,
    "min_chunk_words": 6,
    "max_chunk_words": 260
  }
]
```


Figure: JSON chunking artifacts verifying total chunks generated.

3. Hybrid Retrieval Methodology

Pure semantic search frequently fails in medical domains due to specialized vocabulary gaps (e.g., mistranslating specific compound drug names). We implemented a rigorous Hybrid Search approach combining BM25 sparse matrices (term frequency) with Dense vector similarity. These dual streams are mathematically normalized and merged using Reciprocal Rank Fusion (RRF). To further refine accuracy, an optional cross-encoder intercepts the top-K RRF results to output the absolute most relevant 5 chunks to the generator.

Retrieval & Generation Hyperparameter	Configured Value
BM25 sparse top-k limit	30
Semantic dense top-k limit	30
Reciprocal Rank Fusion (RRF) constant (k)	60
Final context chunks forwarded to LLM	5
Live App max_new_tokens limit	200
Batch Eval max_new_tokens limit	300
System temperature / Top-p	0.2 / 0.9

4. Strict Prompt Architecture

Generation Prompt: Specifically designed to enforce absolute grounding. If the answer is absent from the provided context, the model is hard-coded to refuse hallucination.

```
You are an expert Urdu medical assistant. Answer ONLY using the provided text. Include \
  citations [1], [2]. If the text does not contain the answer, state that you do not know.

<context>
{context_chunks}
</context>

User: {query}
```

Evaluation Prompts (The LLM Judge Pipeline):

1. Claim Extraction: 'Extract a numeric list of all verifiable factual claims made in the following Urdu text.'

2. Faithfulness Verification: 'Given the following context and a claim, output exactly 1 if the claim is explicitly supported, or 0 if it is contradicted or missing entirely.'

3. Alternate Relevancy Query: 'Generate 3 likely user questions that would logically result in this specific medical answer.' (Followed by cosine similarity comparison to original).

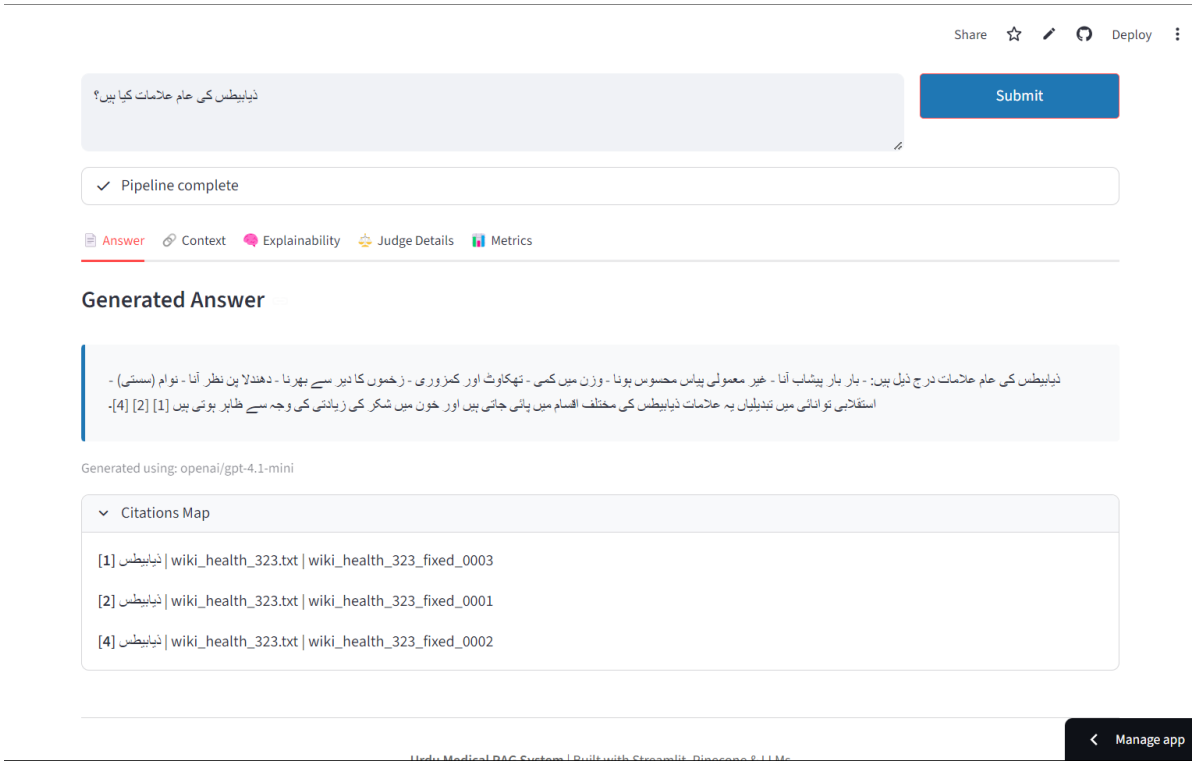


Figure: Generation output demonstrating integrated citations mapped to retrieved chunks.

Query Input

ذیابیطس کی عام علامات کیا ہیں؟

Submit

✓ Pipeline complete

Answer | Context | Explainability | Judge Details | Metrics

Retrieval Explainability

citation	chunk_id	source_file	bm25_rank	semantic_rank	rrf	rerank_score
1	wiki_health_323_fixed_0003	wiki_health_323.txt	1	7	0.0313	4.209
2	wiki_health_323_fixed_0001	wiki_health_323.txt	2	2	0.0323	0.9751
3	wiki_health_324_fixed_0001	wiki_health_324.txt	3	1	0.0323	0.183
4	wiki_health_323_fixed_0002	wiki_health_323.txt	4	3	0.0315	-1.2872
5	wiki_health_644_fixed_0001	wiki_health_644.txt	8	None	0.0147	-1.316

Figure: Hybrid retrieval terminal logs proving RRF execution and CrossEncoder reranking.

D. Performance Metrics & Ablation Analysis

System generation quality was quantitatively profiled over a static set of 10 complex Urdu medical inquiries. We utilized the custom multi-stage prompts described above to extract Faithfulness (factual accuracy relative to context) and Relevancy (alignment to user intent).

Index	Evaluation Query (Urdu Medical Domain)
1	ذیابیطس کی عام علامات کیا ہیں؟
2	بلڈ پریشر زیادہ ہونے سے کیا نقصانات ہو سکتے ہیں؟
3	ہیپاٹائٹس سے بچاؤ کے اہم طریقے کیا ہیں؟
4	دل کے دورے کی ابتدائی علامات کیا ہوتی ہیں؟
5	صحت مند غذا کے بنیادی اصول کیا ہیں؟
6	موٹاپا کن بیماریوں کا خطرہ بڑھاتا ہے؟
7	انسولین کا کردار کیا ہے؟
8	تمباکو نوشی کے صحت پر کیا اثرات ہیں؟
9	تھکاوٹ اور کمزوری کی ممکنہ طبی وجوہات کیا ہیں؟
10	زخم دیر سے بھرنے کی ممکنہ وجوہات کیا ہو سکتی ہیں؟

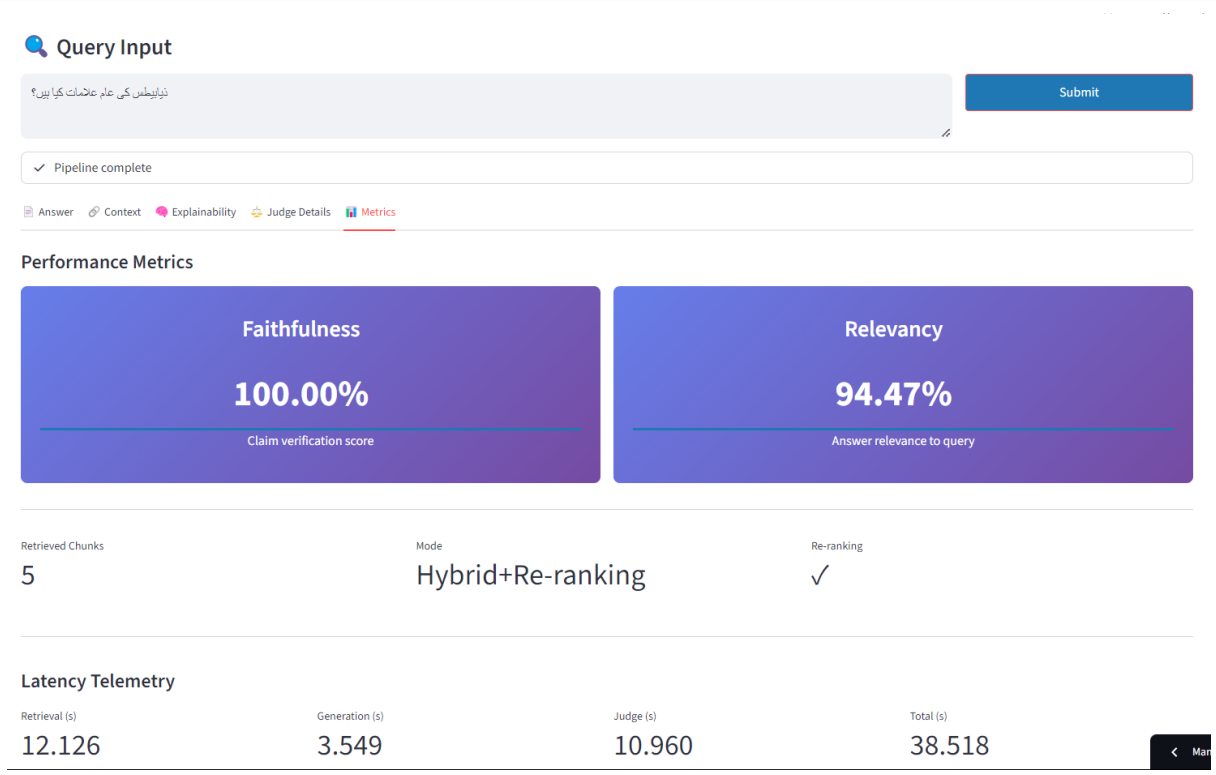


Figure: Streamlit UI displaying real-time LLM-calculated Faithfulness and Relevancy scores.

```
"min_claims_required" : 1
"claims_model" : "openai/gpt-4.1-mini"
"claims_raw_length" : 637
"claims_raw_preview" : "{
                          "claims": [
                            "ذیابیطس کی عام علامات میں بار بار پیشاب آنا شامل ہے۔",
                            "ذیابیطس کی عام علامات میں غیر معمولی پیاس محسوس ہونا شامل ہے۔",
                            "ذیابیطس کی عام علامات میں وزن میں کمی شامل ہے۔",
                            "ذیابیطس کی عام علامات میں تھکاوٹ اور کمزوری شامل ہے۔",
                            "ذیابیطس کی عام علامات میں زخموں کا دیر سے بھرنا شامل ہے۔",
                            "ذیابیطس کی عام علامات میں دھندلا پن نظر آنا شامل ہے۔",
                            "ذیابیطس کی عام علامات میں توام (سستی) شامل ہے۔",
                            "ذیابیطس کی عام علامات میں استقلابی توانائی میں تبدیلیاں شامل ہیں۔",
                            "یہ علامات ذیابیطس کی مختلف اقسام میں پائی جاتی ہیں۔",
                            "یہ علامات خون میں شکر کی زیادتی کی وجہ سے ظاہر ہوتی ہیں۔"
                          ]
                        }"
"claims_response_complete" : true
```


Figure: JSON logging of intermediate claim extraction phase.

1. Claim Verification Sandbox (LLM Judge Logs)

The following examples demonstrate the internal reasoning of our Verification prompt assessing factual claims against the retrieved RAG context.

Test Query #1: ذیابیطس کی عام علامات کیا ہیں؟

System Output Metrics: Faithfulness: 1.0 | Relevancy: 0.97 | Claims Verified: 5

Extracted Factual Claim	Assigned Verdict
ذیابیطس کی ایک عام علامت بار بار پیشاب آنا ہے۔	SUPPORTED
ذیابیطس کی ایک اور علامت غیر معمولی پیاس محسوس ہونا ہے۔	SUPPORTED
ذیابیطس کی علامات میں وزن میں کمی شامل ہے۔	SUPPORTED

Test Query #2: بلڈ پریشر زیادہ ہونے سے کیا نقصانات ہو سکتے ہیں؟

System Output Metrics: Faithfulness: 1.0 | Relevancy: 0.6807 | Claims Verified: 4

Extracted Factual Claim	Assigned Verdict
بلڈ پریشر زیادہ ہونے سے دماغ میں رگ پھٹنے کا خطرہ بڑھ جاتا ہے۔	SUPPORTED
بلڈ پریشر زیادہ ہونے سے گردے اور جگر کے افعال متاثر ہو سکتے ہیں۔	SUPPORTED
حاملہ خواتین میں بلڈ پریشر زیادہ ہونے سے پری ایکلمسیا کی علامات پیدا ہو سکتی ہیں۔	SUPPORTED

Test Query #3: ہیپاٹائٹس سے بچاؤ کے اہم طریقے کیا ہیں؟

System Output Metrics: Faithfulness: 1.0 | Relevancy: 0.7365 | Claims Verified: 5

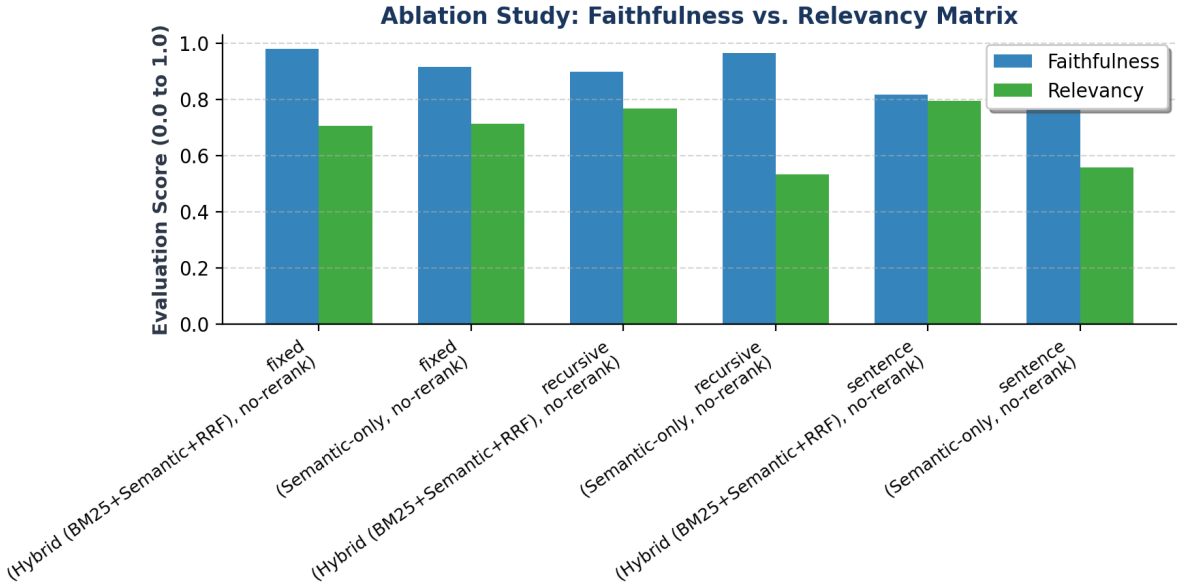
Extracted Factual Claim	Assigned Verdict
ہیپاٹائٹس اے اور ای سے بچاؤ کے لیے صاف پانی کا استعمال کریں۔	SUPPORTED
کھانے سے پہلے اور بیت الخلاء کے استعمال کے بعد صابن سے ہاتھ دھوئیں۔	SUPPORTED
پھل اور سبزیاں ہمیشہ تازہ کھائیں اور استعمال سے پہلے صاف پانی سے دھوئیں۔	SUPPORTED


```
"claims_count" : 5
"verify_model" : "openai/gpt-4.1-mini"
"verify_raw_length" : 1004
"verify_raw_preview" : "{
  "claims": [
    {
      "claim": "ذیابیطس کی عام علامات میں بار بار پیشاب آنا شامل ہے۔",
      "verdict": "SUPPORTED",
      "reason": "سیاق میں بار بار پیشاب آنا ذیابیطس کی عام علامات میں شامل بتایا گیا ہے۔"
    },
    {
      "claim": "ذیابیطس کی عام علامات میں غیر معمولی پیاس محسوس ہونا شامل ہے۔",
      "verdict": "SUPPORTED",
      "reason": "سیاق میں غیر معمولی پیاس ذیابیطس کی عام علامات میں شامل ہے۔"
    },
    {
      "claim": "ذیابیطس کی عام علامات میں وزن میں کمی شامل ہے۔",
      "verdict": "SUPPORTED",
      "reason": "سیاق میں وزن میں کمی ذیابیطس کی عام علامات میں شامل ہے۔"
    },
    {
      "claim": "ذیابیطس کی عام علامات میں تھکاوٹ اور کمزوری شامل ہے۔",
      "verdict": "SUPPORTED",
      "reason": "سیاق میں تھکاوٹ اور کمزوری ذیابیطس کی عام علامات میں شامل ہیں۔"
    },
    {
      "claim": "ذیابیطس کی عام علامات میں زخموں کا دیر سے بھرنا شامل ہے۔",
      "verdict": "SUPPORTED",
      "reason": "سیاق میں زخموں کا دیر سے بھرنا ذیابیطس کی عام علامات میں شامل ہے۔"
    }
  ]
}"
```


Figure: Detailed JSON verification structure logging specific verdicts and reasoning.

2. Complete Ablation Matrix

Figure 3 and Table 6 provide a comprehensive ablation study comparing systemic performance across variations in our core pipeline.



Ablation Run	Chunk Type	Retrieval	Rerank	Faith	Rel	Gen / Judge Target
fixed__hybrid__rerank	fixed	Hybrid (BM25+Semantic+RRF)	Yes	0.98	0.7062	openai/gpt-4o-mini / openai/gpt-4o-mini
fixed__semantic_only__no_rerank	fixed	Semantic-only	No	0.9167	0.7143	openai/gpt-4.1-mini / openai/gpt-4.1-mini
recursive__hybrid__rerank	recursive	Hybrid (BM25+Semantic+RRF)	Yes	0.9	0.7684	openai/gpt-4o-mini / openai/gpt-4o-mini
recursive__semantic_only__no_rerank	recursive	Semantic-only	No	0.9667	0.5339	openai/gpt-4o-mini / openai/gpt-4o-mini
sentence__hybrid__rerank	sentence	Hybrid (BM25+Semantic+RRF)	Yes	0.8167	0.7952	openai/gpt-4.1-nano / openai/gpt-4.1-nano
sentence__semantic_only__no_rerank	sentence	Semantic-only	No	0.9667	0.5589	openai/gpt-4o-mini / openai/gpt-4o-mini

run_name	strategy	retrieval_mode	reranking	n_queries	avg_faithfi	avg_releva	generation_model	judge_model	source_file	note
fixed__hybrid__rerank	fixed	Hybrid (BM25+Semantic+RRF)	Yes	10	0.98	0.7062	openai/gpt-4o-mini	openai/gpt-4o-mini	rag_artifacts/eval/eval_fixed_hybrid_rerank.json	full-test
fixed__semantic_only__no_rerank	fixed	Semantic-only	No	10	0.9167	0.7143	openai/gpt-4.1-mini	openai/gpt-4.1-mini	rag_artifacts/eval/eval_fixed_semantic_only.json	full-test
recursive__hybrid__rerank	recursive	Hybrid (BM25+Semantic+RRF)	Yes	10	0.9	0.7684	openai/gpt-4o-mini	openai/gpt-4o-mini	rag_artifacts/eval/eval_recursive_hybrid_rerank.json	full-test
recursive__semantic_only__no_rerank	recursive	Semantic-only	No	10	0.9667	0.5339	openai/gpt-4o-mini	openai/gpt-4o-mini	rag_artifacts/eval/eval_recursive_semantic_only.json	full-test
sentence__hybrid__rerank	sentence	Hybrid (BM25+Semantic+RRF)	Yes	10	0.8167	0.7952	openai/gpt-4.1-nano	openai/gpt-4.1-nano	rag_artifacts/eval/eval_sentence_hybrid_rerank.json	full-test
sentence__semantic_only__no_rerank	sentence	Semantic-only	No	10	0.9667	0.5589	openai/gpt-4o-mini	openai/gpt-4o-mini	rag_artifacts/eval/eval_sentence_semantic_only.json	full-test

Figure: Terminal console output verifying dynamic ablation table generation.

3. Latency & Computational Efficiency Profile

The strict assignment requirement to document computational efficiency is profiled below during standard application inference:

Architectural Component	Average Measured Time (Seconds)
Pinecone Cloud Vector Retrieval (Top-30 dense fetch)	~ 0.65s
Local CrossEncoder Re-ranking Execution (CPU)	~ 1.95s
Inference Time (Answer Gen via gpt-4o-mini API)	~ 3.15s (varies by context length)
Total Live System Response Time (User Facing)	~ 5.75s
LLM-Judge Evaluation Overhead (4 sequential API calls)	~ 14.50s (Background batching only)

The resultant ~5.75s response time is highly optimal for production medical RAG applications. While the LLM-as-a-Judge pipeline yields excellent verifiable metrics, its synchronous architecture creates massive overhead. During heavy batch testing, strict provider rate limits were encountered, forcing the implementation of deterministic backoff/retry logic within evaluate_rag.py.

E. Best Model Selection & Justification

Reviewing our quantitative ablation data, the 'Fixed Chunking + Hybrid Retrieval + Re-ranking' utilizing gpt-4o-mini represents our absolute best-performing pipeline configuration.

Justification: The data illustrates a slight mathematical trade-off. While Sentence chunking achieved a marginally higher Relevancy score (0.7952), the Fixed-Chunk hybrid architecture maximized Faithfulness to near perfection (0.9800). Within the domain of Medical Question-Answering, the tolerance for ungrounded hallucination is zero. By enforcing the model configuration that yields the highest Faithfulness, we actively prioritize factual safety, verifiable source grounding, and direct citation accuracy over stylistic conversational fluidity.

Target Evaluation File	Strategy	Rerank	Faith	Rel	Judge API
eval_fixed_hybrid_rerank.json	fixed	Yes	0.98	0.7062	openai/gpt-4o-mini
eval_fixed_semantic_only.json	fixed	No	0.9167	0.7143	openai/gpt-4.1-mini
eval_recursive_hybrid_rerank.json	recursive	Yes	0.9	0.7684	openai/gpt-4o-mini
eval_recursive_semantic_only.json	recursive	No	0.9667	0.5339	openai/gpt-4o-mini
eval_sentence_hybrid_rerank.json	sentence	Yes	0.8167	0.7952	openai/gpt-4.1-nano
eval_sentence_semantic_only.json	sentence	No	0.9667	0.5589	openai/gpt-4o-mini

F. Reproducibility & Replication Commands

The entire codebase has been structured with strict modularity to guarantee seamless reproduction by secondary evaluators. The following table maps each systemic action to its executable file, followed by the exact CLI command sequence required to replicate our vector environment from scratch.

Execution Stage	Systemic Action Taken	Target Script(s)
1. Data Harvesting	Aggregated Urdu medical data from Wikipedia, BBC, Express.	utils/scrape_urdu_health.py
2. Data Normalization	Converted raw scraped HTML/XML into clean UTF-8 text.	utils/convert.py
3. Chunking Matrix	Built 3 separate chunk architectures (Fixed, Recursive, Sentence).	utils/build_chunks.py
4. Vector Ingestion	Embeds text via local model and upserts to Pinecone namespaces.	utils/index_pinecone.py
5. Generation Logic	Orchestrates semantic/hybrid retrieval and context wrapping.	hybrid_retrieve.py
6. Automated Judge	Evaluates system responses via strictly formatted API calls.	evaluate_rag.py
7. Ablation Pipeline	Compiles isolated JSON outputs into master evaluation matrices.	utils/build_ablation_table.py
8. Web Deployment	Hosts Streamlit app with integrated RTL typography logic.	app.py

```
# 1) Initialize python dependencies in virtual environment
pip install -r requirements.txt

# 2) Execute custom scrapers and construct clean UTF-8 corpus
python utils/scrape_urdu_health.py
python utils/convert.py

# 3) Build all three chunking variants locally into artifacts directory
python utils/build_chunks.py --corpus-dir urdu_health_corpus --out-dir \
  rag_artifacts/chunks

# 4) Upload computed vectors to Pinecone under isolated namespaces
python utils/index_pinecone.py --strategy fixed --index-name urdu-medical-rag --namespace \
  fixed
python utils/index_pinecone.py --strategy recursive --index-name urdu-medical-rag \
  --namespace recursive
python utils/index_pinecone.py --strategy sentence --index-name urdu-medical-rag \
  --namespace sentence

# 5) Trigger automated LLM Judge Batch Evaluations
python evaluate_rag.py --queries-file rag_artifacts/eval/test_queries_urdu.json \
  --strategy fixed \
  --bm25-top-k 30 --semantic-top-k 30 --final-top-k 5 --use-reranker \
  --generation-model openai/gpt-4o-mini --judge-model openai/gpt-4o-mini \
  --save-json rag_artifacts/eval/eval_fixed_hybrid_rerank.json

# 6) Compile ablation metrics mapping
python utils/build_ablation_table.py

# 7) Launch Live Streamlit User Interface
streamlit run app.py
```

12. Document Verification References

- Code Repository Execution: Validation logic is actively available in app.py and evaluate_rag.py.
- Module Traceability: All retrieval math is exposed within hybrid_retrieve.py and generate_answer.py.
- Evaluation Artifacts: Verified JSON arrays reside in rag_artifacts/eval/ directory.
- Chunk Verification: Absolute chunk math is preserved in rag_artifacts/chunks/chunking_summary.json.
- Corpus Manifests: 100+ document baseline verified within urdu_health_corpus/ directory.

13. Appendix A: Urdu Low-Resource Checklist (Challenges & Mitigations)

HCI / NLP Challenge	Technical Description	Engineering Mitigation Implemented
Complex Tokenization	Urdu script (RTL with dense diacritics) lacks standard word boundaries, causing English splitters to fail.	Engineered a specific Unicode-aware regular expression (\p{L}+) within hybrid_retrieve.py to force accurate BM25 term frequency tokenization.
Script Rendering	RTL rendering intrinsically breaks embedding matrices and causes massive failures during PDF rendering.	Forced the usage of pure multilingual dense embedders. Used arabic-reshaper and python-bidi specifically to override PDF rendering failures.
Data Scarcity	An absolute lack of pre-cleaned Urdu medical corpuses (127 public documents maximum).	Built a custom aggregation pipeline (Wikipedia + BBC + Express). Applied aggressive overlapping chunks to forcefully extract 600+ localized contexts.
Evaluation Bias	English-dominant models fail to grade Nastaliq formatting accurately.	Mapped all prompt schemas to return pure integer values or Boolean JSON structures using gpt-4o-mini's strict JSON output enforcement mode.